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1. **SCOPE OF RADIATION SAFETY**

   a. The Hospital Radiation Safety program applies to all locations where radiation-producing machines are used or stored, regardless of ownership or the location.

   b. It applies to all persons working at or frequenting these locations, regardless of their relationship with the Hospital

   c. It applies to all radiation-producing machines at these locations, regardless of ownership of the machines.

2. DEFINITIONS & ABBREVIATIONS

Definitions

a. **Absorbed Dose**
   i. The amount of energy imparted to matter by ionizing radiation per unit mass of irradiated material. The unit of absorbed dose is the Gray (Gy)

b. **ALARA**
   i. Acronym for As Low As Reasonably Achievable.

c. **Calibration**
   i. The check or correction of the accuracy of a measuring instrument to assure proper operational characteristics

d. **Critical Organ**
   i. The organ or tissue, the irradiation of which will result in the greatest hazard to the health of the individual

e. **Declared Pregnant Worker**

i. A woman who has voluntarily informed her employer, in writing, of her pregnancy and the estimated date of conception

f. **Dose Rate**
   i. The radiation dose delivered per unit of time

g. **Dosimeter**
   i. A portable instrument for measuring and registering the total accumulated exposure to ionizing radiation

h. **Geiger-Mueller (G-M) Counter**
   i. Radiation detection and measuring instrument.

i. **Gray**
   i. The international (SI) unit of absorbed dose in which the energy is equal to one Joule per kilogram

j. **Half Value Layer**
   i. The thickness of any specified material necessary to reduce the intensity of an x-ray or gamma ray beam to one-half its original value

k. **Inverse Square Law**
   i. The intensity of radiation at any distance from a point source varies inversely as the square of that distance

l. **Ionizing Radiation**

i. Any radiation capable of displacing electrons from atoms or molecules, thus producing ions

m. **Occupational Radiation Dose**
   i. The dose received by an individual in the course of employment.

n. **Radio sensitivity**
   i. The relative susceptibility of cells, tissues, organs, organisms, or other substances to the injurious action of radiation

o. **Sievert**
   i. The international (SI) of dose equivalent.

p. **Thermo luminescent Dosimeter (TLD)**

i. Crystalline materials that emit light if they are heated after they have been exposed to radiation

1.1. **Abbreviations:**

a. **AERB :** Atomic Energy Regulatory Board

b. **BARC:** Bhaba Atomic Research Center

c. **ALARA:** As Low As Reasonably Achievable

d. **RSO:** Radiation Safety Officer

e. **GM:** Geiger-Muller Counter

f. **Gy:** Gray

g. **Pb:** Lead

h. **mSv:** millisievert

i. **TLD:** Thermo luminescent Dosimeter

j. **NOC:** No Objection Certificate

k. **SSD:** Source-Skin Distance

l. **THF:** Tuned high frequency

2. **Radiation Safety Policies:**

Statutory Requirements:

Policy:

The Atomic Energy Regulatory Board (A.E.R.B) is entrusted with the responsibility of developing and implementing appropriate regulatory measures to ensure radiation safety. Statutory requirements with regard to radiation safety are as follows:

1. Commissioning and Decommissioning of X-ray Equipment has to be registered with AERB.
2. Direct assistance to the patient while being X-rayed has to be avoided. If assistance is required, appropriate precautions have to be taken by the person who will assist by making use of appropriate protective material and devices which are available.
3. Fetal protection measures to be used.
4. Periodic inspection of X-ray equipment and shielding features is conducted regularly.
5. Personnel monitoring facility be provided to all radiation workers.
6. Presence of uninvolved staff, patients and persons in any X-ray room must be avoided.
7. Regular maintenance and calibration of the unit must be carried out.
8. Reproductive organs must be particularly shielded.
9. Services of qualified radiologists and X-rays technologists to be used.
10. Servicing and calibration of X-rays equipment should be undertaken by qualified, trained and authorized service engineer.
11. There should be transfer of radiographs and reports to avoid repeated X-rays examinations.
12. X-rays equipment meeting design certification and type approval requirement by AERB only shall be used.
13. X-ray examination of pregnant women and children should be avoided as far as possible.

Personal Protective Equipments:
Policy:

1. No person shall operate or permit the operation of certified or uncertified medical radiographic equipment systems unless the following conditions are met:

a. Only individuals required for the medical procedure, for training or for equipment maintenance shall be in the radiographic room during an exposure.
    i. Individuals who are present in a radiographer room during any exposure shall wear protective aprons of at least 0.25 mm lead equivalent during every exposure.

b. When a patient must be provided with auxiliary support during a radiation exposure and
Mechanical holding devices are insufficient; the following procedures shall be followed:
    i. The person holding the patient shall be protected with a lead apron of at least 0.25 mm lead equivalent;
    ii. The person holding the patient shall be protected with lead gloves of at least 0.25 mm lead equivalent if the hands must be placed in the useful beam.
    iii. Radiographers not to hold the patient during a radiation exposure, except in a life-threatening situation.
    iv. No person shall be employed, routinely assigned, or required to hold a patient during radiographic and fluoroscopic procedures;
    v. If a patient must be held during the x-ray exposure, non-radiation workers such as nurses or members of the patient's family may be asked to perform this duty.

c. Gonad shielding of not less than 0.5 mm lead equivalent shall be used on a patient during Radiographic procedure, except for cases in which this would interfere with Diagnostic procedure.

d. The operator shall collimate x-ray beam limitation to ensure that the x-ray field does not extend beyond the Region of interest.
e. The Radiographic field shall be restricted to the areas of clinical interest as far as practical;
f. A method to observe the patient during the x-ray exposure (Lead glass) shall be provided for all units.
g. During radiographic exposure, the operator shall stand behind the protective barrier.

h. The department incharge shall provide safety rules to each individual operating x-ray equipment including any restrictions as to the operating technique required for the safe operations of the particular x-ray apparatus, and require that the operator sign a form acknowledging that the safety manual was read.
i. No person shall permit or arrange for the intentional irradiation of a human being except for the purpose of medical diagnosis or treatment;
j. No person shall deliberately expose an individual to the useful beam for the sole purpose of training or demonstration.
k. No person shall operate an ionizing –radiation-producing machine unless that person understands and uses the principles of radiation safety to keep radiation exposure as low as reasonably achievable (ALARA).

List of Personal Protective Equipment:

Safety Guidelines:

Policy:

1. All safety guidelines shall be observed.

2. Report every injury, no matter how slight, to your in charge.
3. No intoxicating liquor shall be consumed while on duty. Anyone who is found under the influence of alcohol or drugs will be suspended as per rule.
4. Keep fit for your job, eat properly and get sufficient rest to meet the demands of your job.
5. Take a special interest in the new or inexperienced persons and help them with the small details of the job.
6. Be sure to notify all persons of any dangerous situations that might affect your work area.
7. Remember the patient; never leave him/her unattended.
8. Know all the hospital emergency codes and be sure of your responsibilities.
9. When dealing with the extremely large patient, be sure to seek help and lift the patient correctly.
10. Know your fire extinguishers, their locations and the use.
11. Use good housekeeping techniques at all times.
12. Remember the department security.

General Radiation Protection :

Policy:

1. A qualified radiographer must only do all radiographic techniques and procedures. All radiographers must take necessary steps in reducing radiation dose to the patient.
2. Check the correct patient for correct examination.
3. Plan your technique to reduce the radiation dose
4. Close the X-ray room door properly and tightly.
5. Provide the necessary radiation protection.
6. Collimate the radiation beam to necessary area only.
7. Give proper and correct instructions.
8. Select the appropriate exposure factor.
9. Place the correct ID Permanent Number for correct patient
10. Avoid unnecessary repeats.
11. For female patients check whether they are pregnant.
12. Limit number of people in the X-ray room while X-ray is being done.
13. Mobile X-ray request only if it is necessary.
14. All staffs must wear radiation-monitoring badge while in the radiology department.
15. Use only high-speed cassette (Green Sensitive) - to reduce exposure.
16. Clear all staffs from room during Mobile X-ray /Provide Lead apron to the next Bed patient if he he/she is not able to move.
17. Everybody should be 6 feet away from x-ray tube during Mobile x-ray

Departmental Policy :

Policy:

1. Incharge of radiology services is responsible for maintaining safety standards, developing safety rules and supervising and training personnel in departmental standards.
2. Incharge of radiology services is responsible for notifying the Management in case of any safety hazard.
3. All radiology employees shall report defective equipment, unsafe conditions, acts or safety hazards to Manager of radiology services.
4. Keep electrical cords clear of passage ways. Do not use electrical extension cords without prior informing the facility department.
5. All equipments and supplies must be properly stored.
6. Scissors, knives, pins, razors blades and other sharp instruments must be stored and used safely.
7. All electrical machines, with heat producing elements, must be turned off or unplugged when it is in not use.
8. Smoking is prohibited, per hospital smoking policy.
9. Do not permit rubbish to accumulate.
10. Notify the facility department immediately of illumination and Air conditioning effect Problems.
11. Furniture and equipment must be allowed adequate passage and access to exits at all times.
12. Employee who discovers the spill should inform minor spills, such as water & chemical spill , to Housekeeping team. This shall be done immediately.
13. Report faulty equipment to the Biomedical Engineer or vendor, per policy.
14. Obey warning signs.

15. File drawers and cabinet doors shall be closed when not in use.
16. Wear suitable clothing, only authorized personnel shall be allowed in X-ray room.

Radiographers contribution to the safety programme :

Policy:

1. It is the responsibility of the RADIOGRAPHER to have thorough knowledge and apply on the job instructions for all personnel regarding safe practices.
2. Department incharge is responsible for the degree to which his/her personnel have gained the knowledge and skills necessary to perform safely and effectively in their particular position.
3. Individual departments will establish and publish safe work rules which reduce accident probability. Development of these rules should involve:
    a. A review of all work methods and practices
    b. A review of all past accident experiences
    c. Recommendations by supervisory personnel
    d. Recommendations by personnel
    e. Investigate personnel injuries within the department
    f. Cooperate with the Department Incharge in the promotion of this activities
    g. Assist in monitoring Safety Recommendations.

Patient Safety

Policy:

1. All patients shall receive the utmost care and attention
2. No patients shall be left unattended.
3. Appropriate Personal protective equipment shall be kept available at all times, personnel shall be trained in their use.
4. One staff personnel will attend all trolley patients.
5. When the lift is used for any patient, one staff must be present on the lift.
6. All wheelchair will have wheel locked when the wheelchair is on the lift and while the patient is entering or existing the wheelchair.
7. When a trolley is used for the transportation of the patient, the side rails will always be up. Trolley wheel must be locked while trolley is on lift.
8. Transportation method for out patient will be based on status of the patient when assessed by the hospital staff prior to x-ray/scanning. An out patient determined to be ambulatory will be allowed to walk to the unit.
9. Ambulatory patients are to be accompanied from the radiology department to the MRI room and back to the department.
10. The Bed roll of U.S.Scan Table shall be changed for each patient to prevent any kind of cross Infection/contamination.
11. X-ray bed roll will be provided for Infectious patients,
12. Any of the hospital employees who knowingly disguard the patients' safety by disobeying the aforementioned policies will be subject to disciplinary action by management.

13. When using portable X-ray units, only radiographer and the patient shall be in the room at the time of exposure. The door shall be closed and the radiographer shall stand at least six feet from the portable unit.
14. All X-ray switches shall be in allocation where they cannot be accidentally energized.
15. TLD shall be worn by all Staffs during working hours in department. Badges will be processed and recorded Quarterly.
16. A routine check shall be made X-ray equipment before using. Recalibrate when tubes are changed or machines modified.
17. If a patient must be held during x-ray, the assistant holding the patient will were a lead-lined apron during the entire procedure.
18. Nurse or Radiographer who calls for patient will check the correct identify of the patient.

19. The radiographer shall always stand in the lead line control cubicle when making an exposure.
20. The X-ray tube shall never be pointed directly towards the control cubicle, In unavoidable circumstances to collimate accurately and wear lead apron and follow ALARA technique.
21. The doors to the X-ray room must always be kept closed.
22. Return equipments to its proper location when not in use.
23. Do not obstruct fire equipment. Know location of fire-fighting equipment and how to use it. Know evacuation routes and what to do in case of fire.
24. Patients such as a children and pregnant women shall be shielded.

25. The radiographer will ensure that all the Infant/children being radiographed have proper shielding and proper collimation of the X-ray beam to expose only the required Anatomy.
26. All expectant females if necessary to be X-rayed will be properly shielded and the X-ray beam collimated to the area of interest only.
27. Pregnant females will not be permitted in the X-ray room during exposure.

Instruction guidelines for proper operation of X-Ray producing equipments :

Policy:

1. Appropriate personal protective equipment (Radiation Protective devices) is to be worn where there is a risk of Radiation exposure.
2. Personnel monitoring devices should always be worn when working with radiographic equipment. The devices worn should be those issued for the current time period and should be worn under the lead apron. Those workers wearing TLD badges should ensure that the Card has been properly inserted into the Cassette holder.
3. Only persons whose presence is necessary should be in the radiographic room during exposure. All such persons who are subject to direct scatter radiation shall be protected by aprons or whole body protective barriers of not less than 0.25 mm lead equivalent.
   Note: A lead apron (Pb) of 0.25 mm lead equivalence will reduce scattered x-rays by 95%.

4. Mechanical supporting or restraining devices shall be used when a patient or film must be held in position for radiography . If a patient must be held by an individual, that individual shall be protected with appropriate shielding devices of at least 0.25 mm lead equivalence for whole body protection and at least 0.5 mm lead equivalence for any part of the holder's body that is exposed to the primary x-ray beam.
5. The x-ray beam should always be collimated to the smallest area consistent with clinical requirements and should always be aligned accurately with the patient and film.
6. Mobile equipment should be used only for examinations where it is impractical to transfer patients to permanent radiographic installations.
7. The operator should stand behind the barrier provided for his/her protection during radiographic exposures at permanent radiographic installations and should stand as far as possible (at least 6 feet) from the patient when operating the mobile equipment.
8. Each mobile radiographic equipment operator, prior to making an exposure, should ask anyone within 6 feet of the x-ray tube and/or patient being radiographed to move further away until the exposure is complete. Those persons who must remain within 6 feet of the patient and/or x-ray tube must be protected by whole body aprons or barriers of at least 0.25 mm lead equivalence. The operator shall give an audible warning before the exposure is made.
9. When Making X-ray exposures, it is advisable to place the image intensifier closest to the region of interest. This results in better image quality and reduces risk from potential hazards.
10. Special precautions, consistent with clinical needs, should be taken to minimize exposure of the embryo or fetus in patients known to be or suspected of being pregnant.
11. No abdominal area radiographic imaging shall be performed on a pregnant or potentially pregnant patient without the approval of a qualified physician. If the x-ray procedure does include the abdominal region of the pregnant or potentially pregnant patient, the

examination shall not be performed without approval from a diagnostic radiologist. Although it is the responsibility of the referring physician to determine pregnancy status, those operating diagnostic x-ray equipment will ask all patients of childbearing age whether or not they are pregnant and the date of their last menstrual period. This information is to be recorded on the study requisition prior to examination.

12. If the x-ray procedure does not include the abdomen or pelvis of the pregnant or potentially pregnant patient, the abdominal region should be shielded with at least 0.25 mm lead equivalence, and the examination performed without regard to pregnancy.
13. The minimum source-skin distance (SSD) for all mobile radiographic x-ray units must be 30 centimeters.
14. The radiation protection program is guided by the concept of keeping radiation exposure As Low as Reasonably Achievable (ALARA).
15. Remember that radiation cannot be seen or felt, but can be detected with radiation survey meters.
16. Radiation exposure of all individuals routinely working with sources of radiation is monitored with a TLD (Thermo luminescent dosimeter) badge. The devices are checked quarterly.
17. Radiation exposure can be minimized by utilizing three basic principles
    a. **Time:** Shorter exposure time means a lower dose.
    b. **Distance:** Doubling the distance from a radiation source means one-fourth the dose rate.

       Tripling the distance gives one-ninth the dose rate.

       Inverse square law.
    c. **Shielding:** The use of appropriate shielding greatly reduces the dose rate.

Ultrasound Scan Safety Guidelines:

Policy:

1. **Patient safety:**

   Always include proper identification with all patient data and verify the accuracy of the patient's name or ID numbers when entering such data. Make sure correct patient ID is provided on all recorded data and hard copy prints.

a. Use condoms when using Transvaginal probe and scanning infectious patients

b. Discard condoms safely.

2. **Diagnostic Information:**

   Equipment or incorrect settings ca result in measurement errors or failure to detect details within the image.

3. **Mechanical Hazards:**

   Damaged probes or improper use and manipulation can result in injury or increased risk of infection.

4. **Electrical Hazard:**

   A damaged probe can also increase the risk of electrical shock if conductive solutions come in contact with internal live parts. Inspect probes often for cracks or openings.

Lead Apron inspection /radiation protective device

Policy:

1. Lead aprons must be used to protect staffs and patients from unnecessary radiation exposure from diagnostic radiology procedures.
2. Health care organization must perform inspections on medical equipment, including lead aprons, lead protective devices etc.
3. **Inspection Frequency**
   a. **Monthly:** The Inspection consists of visual check to look for obvious tears, cuts, or etc.
   b. **Bi annually:** Apron must be placed on the table and checked using the automatic brightness control (fluoroscopy method)

4. Aprons must be stored properly in hangers.
5. Do not fold or pile up.
6. Check for cracks.

Handling of Chemicals:

Policy:

1. Most chemicals are harmful to some degree. Avoid direct contact with any chemical.
2. Wash thoroughly with hand wash solution and water whenever a chemical contacts your skin.
3. Never taste or smell a chemical.
4. All container and chemicals must be labeled clearly. Do not use any substance in an unlabeled container.
5. Always pour concentrated solutions slowly into water or into less concentrated solution while stirring. Always wear safety spectacles and Plastic apron & Mask while diluting solutions.
6. Keep flammable solvents away from heat and sunlight. Do not heat flammable solvents directly over a naked flame or hot plate.
7. Discard safely
8. Wear mask during preparing fresh and filling/ discarding processing solutions.

Chemical Waste Disposal:

Policy:

1. This Fixer & Developer is recommended for processing Medical X-ray films in automatic Film processor equipment.
2. **Storage**
   a. The chemicals must be stored and used at 4 to 29°C.

b. Discard if there is evidence of contamination, dirt, over-dilution, excessive evaporation, or crystallization safely.

3. **Mixing Instructions**
   a. Instructions for mixing replenish and/or working solutions are provided by leaflets for individual size packages.
4. Disposal: -Developer: Used Developer should be neutralized (pH 7-9) and flushed with large quantities of water to the sewer system.
   a. UNUSED developer contains hydroquinone which is a toxic substance, so unused developer cannot go down the drain.
   b. Keep developer and used fixer separated.
   c. If used fixer and developer accidentally get mixed together, the mixture must be disposed of as dangerous waste.
   d. Disposal: Fixer - Used fixer from X-ray processing is defined as a Hazardous waste because it contains high concentrations of silver.
   e. Collect used fixer in a container marked “Hazardous Used fixer”. Keep fixer separately- storage capacity 200 Liters.
   f. The supplier will be asked to take it back at cost. Keep disposal receipts.
   g. After sale close all the lids of the sold fixer container seal it with plaster to avoid spillage during transportation
   h. The films are disposed with used fixer to the hypo buyer as solid waste.